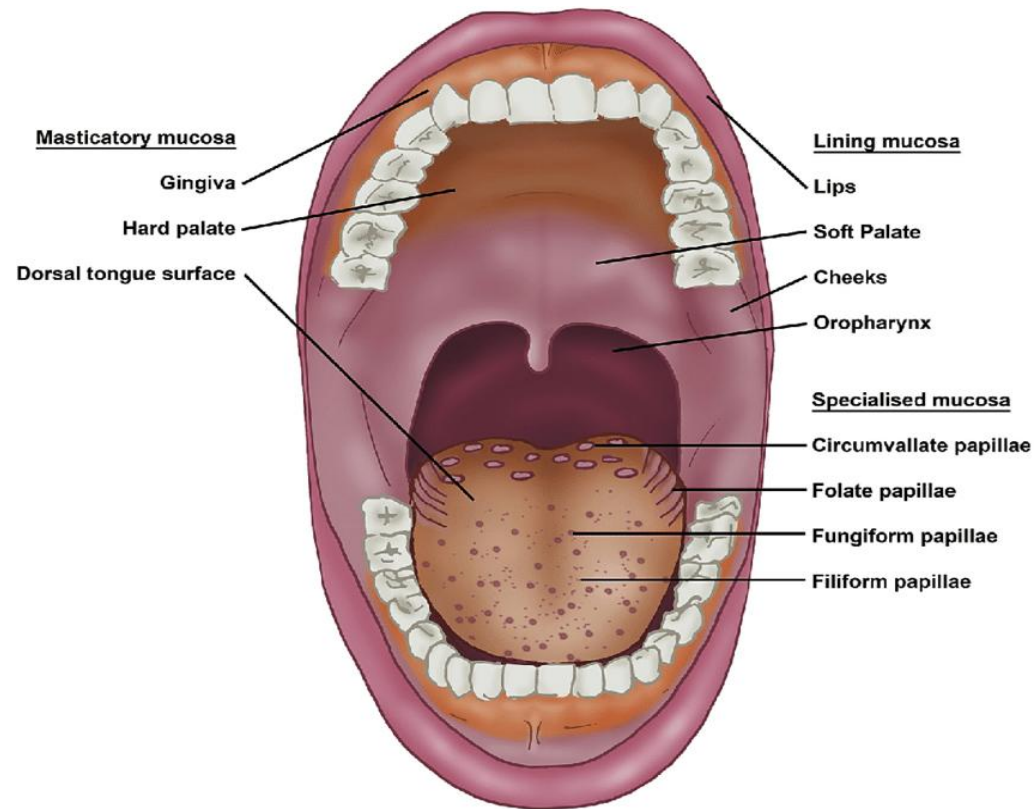
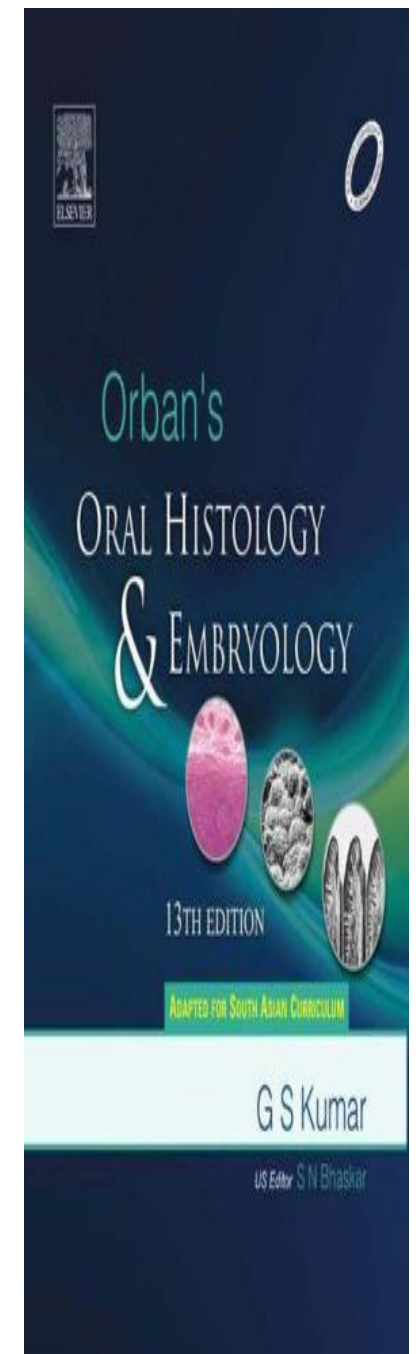


ORAL MUCOUS MEMBRANE-1



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ORAL CAVITY IN HEALTH

*AS YOU KNOW MORE
YOU GROW MORE..*



Lecture Learning Outcomes: L-11 OMM-1

By the end of this lesson students will be able to

11.1: Define oral mucous membrane and Classify the oral mucous membrane.

11.2: Describe the basic structure including histology of the oral mucosa.

11.3: Discuss the difference between skin and oral mucosa.

Definition of oral mucosa or oral mucous membrane (OMM)

MUCOUS MEMBRANE:

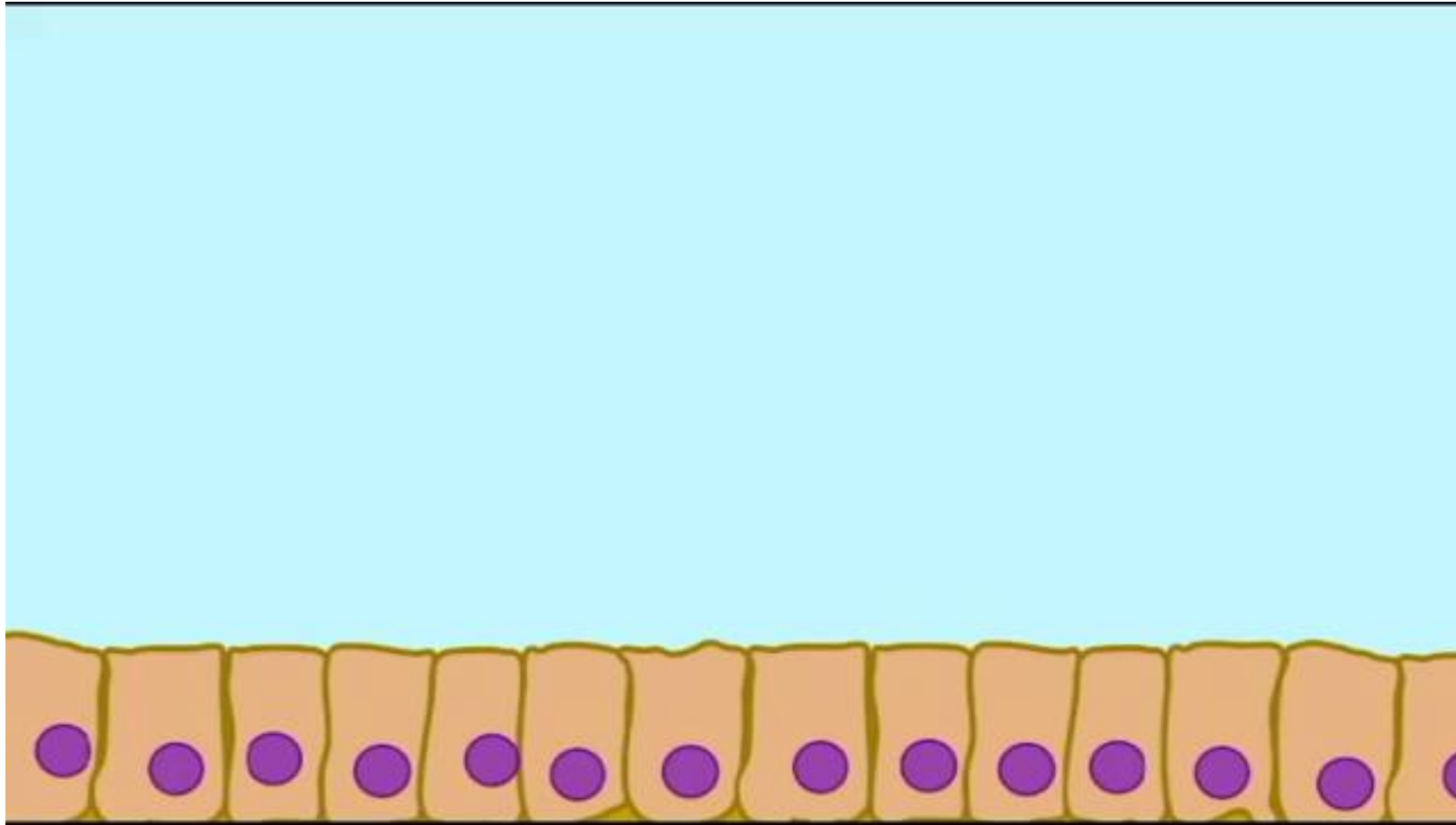
Moist lining of the gastrointestinal tract, nasal passages and other body cavities that communicates with the exterior of the body .

ORAL MUCOUS MEMBRANE

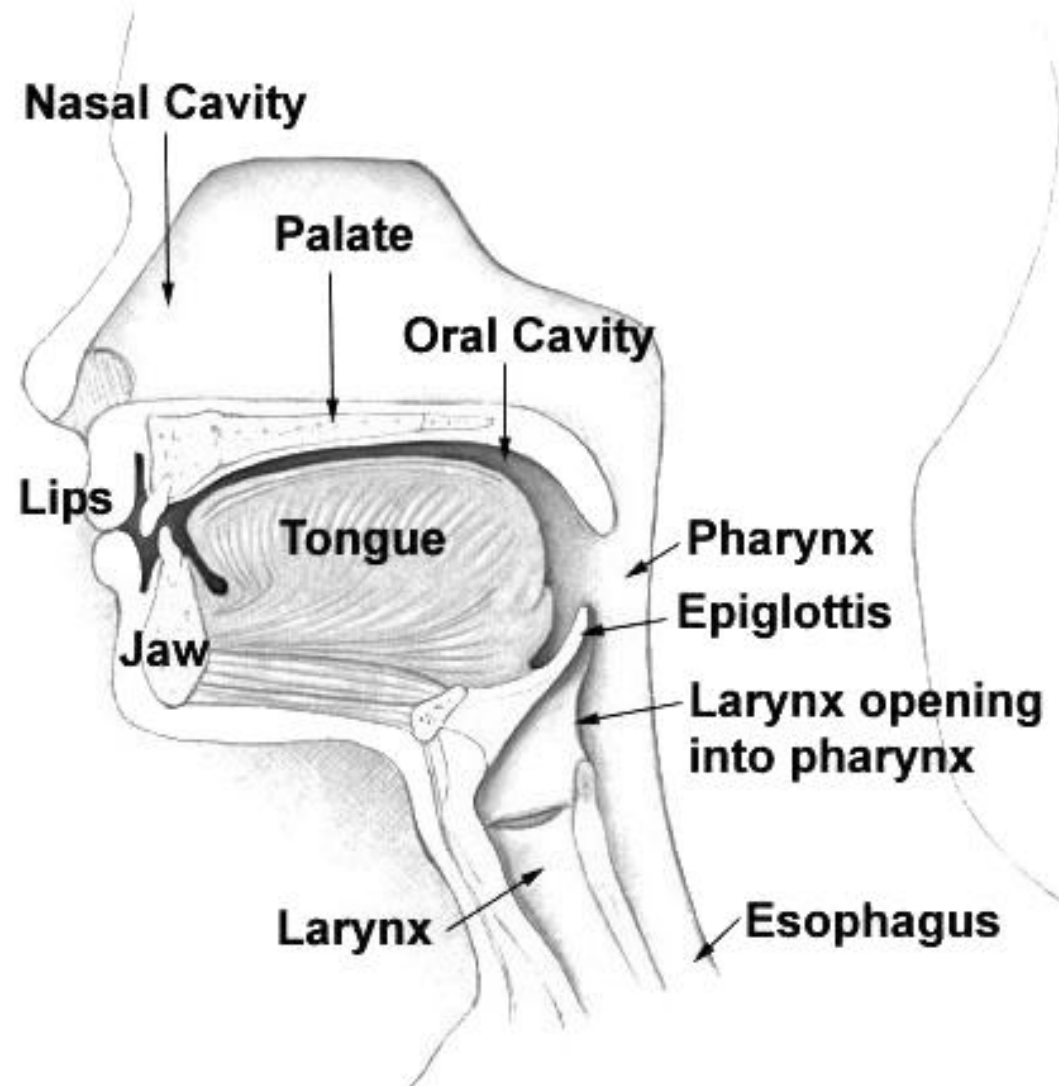
In the **oral cavity the lining** is called as oral mucous membrane or oral mucosa.

Its maintain structural integrity by a process of continuous renewal.

Shedding or desquamation of old cells at the superficial cell layer.

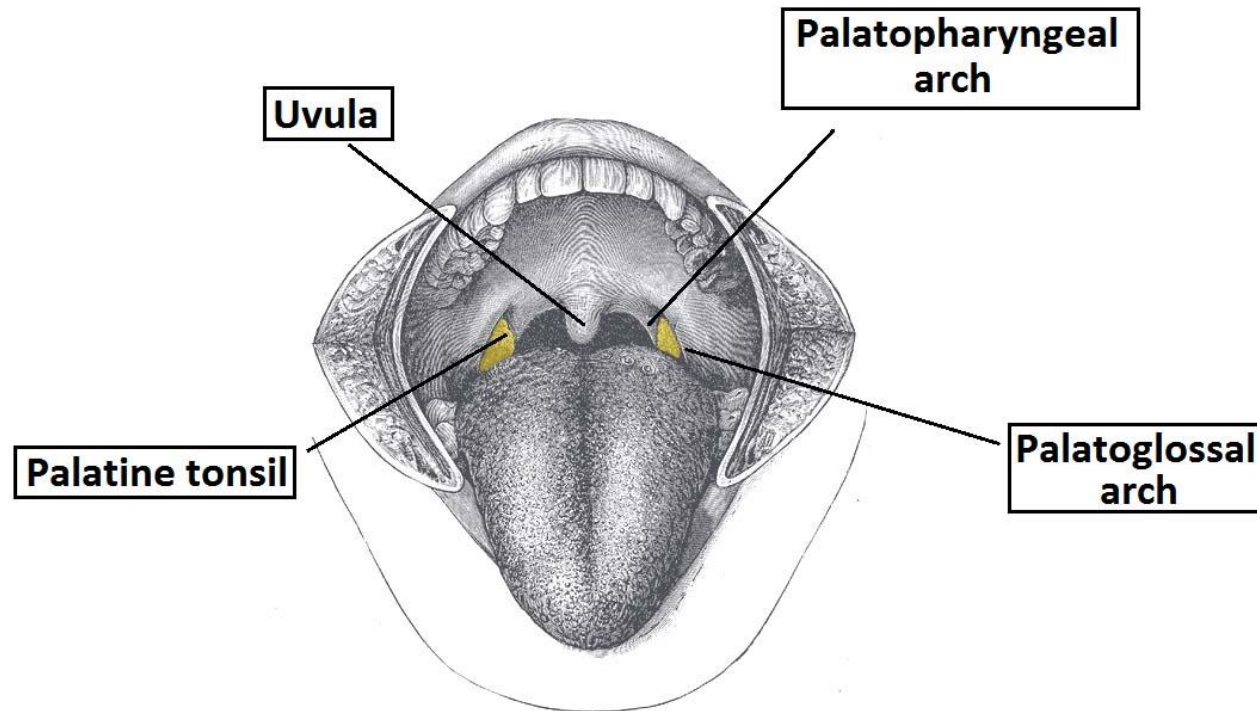


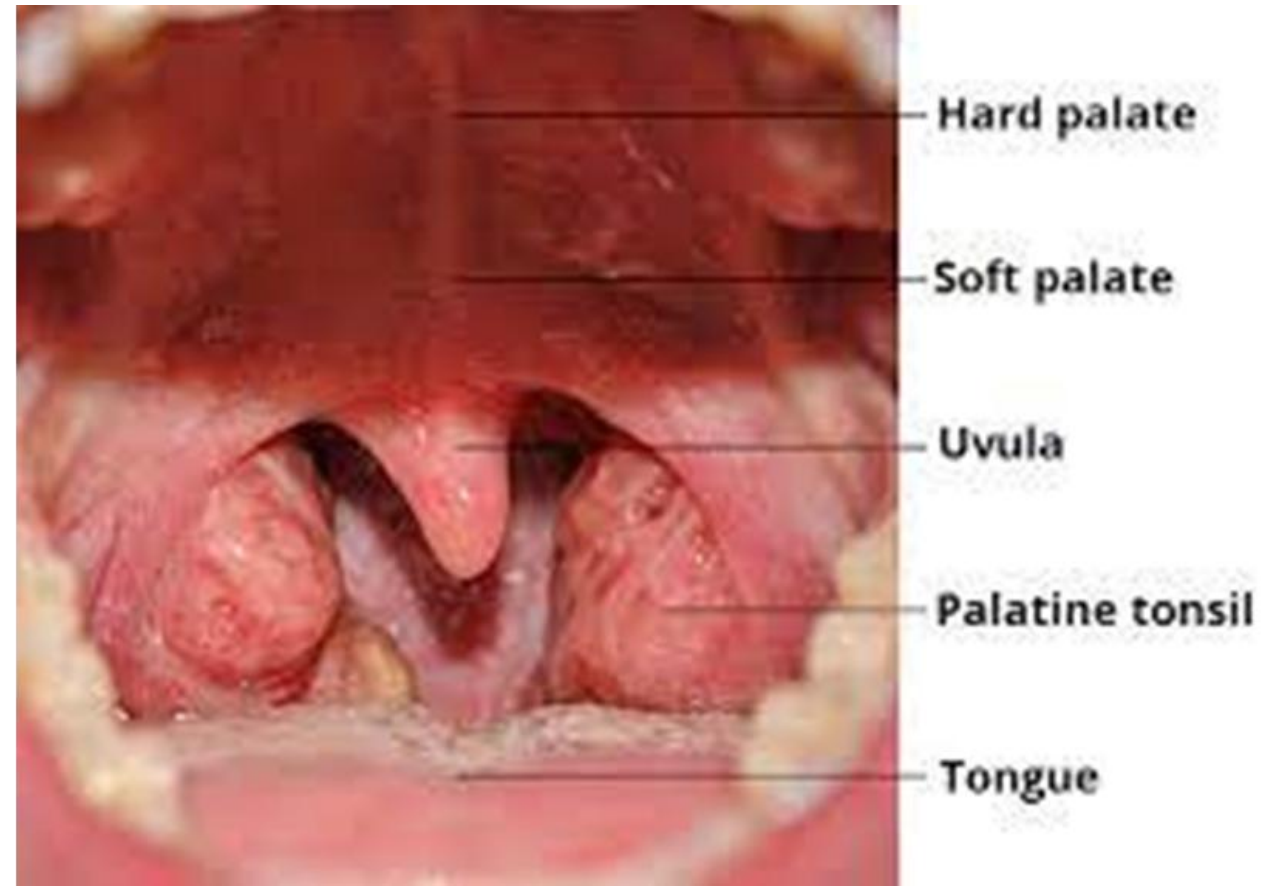
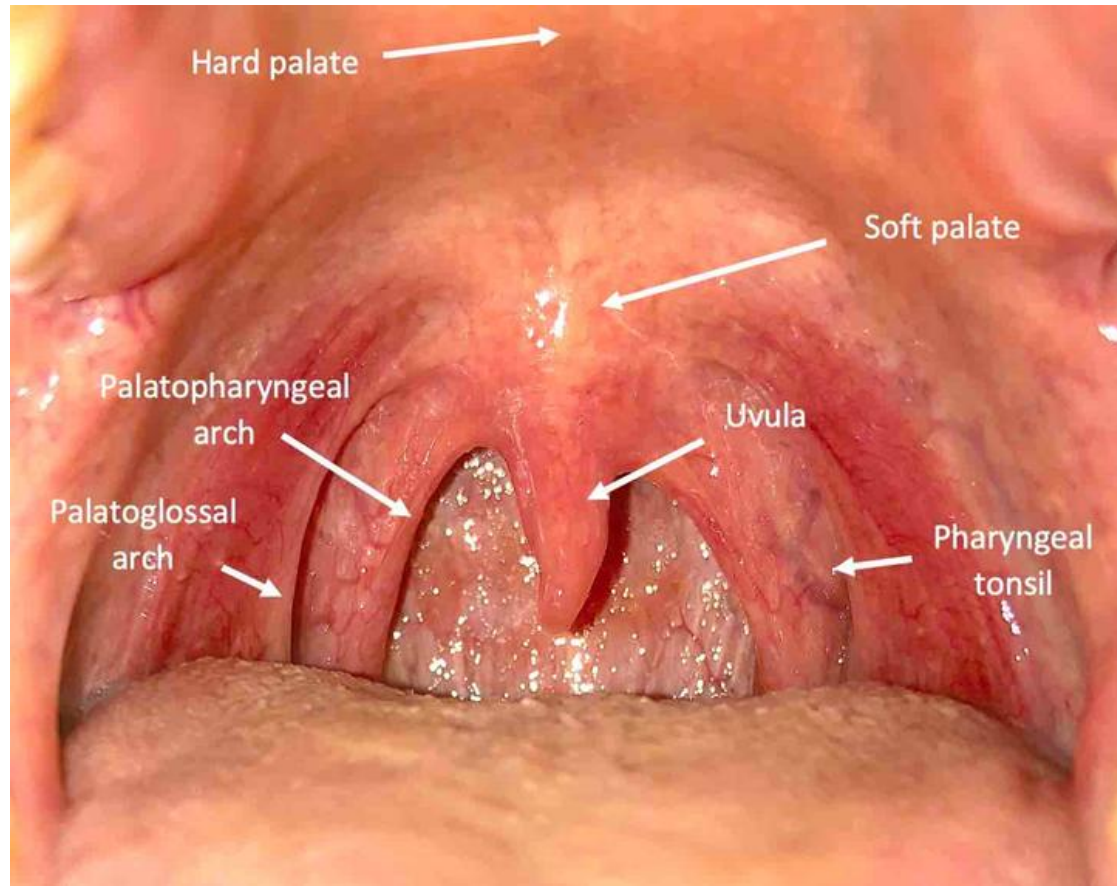
ORAL MUCOUS MEMBRANE

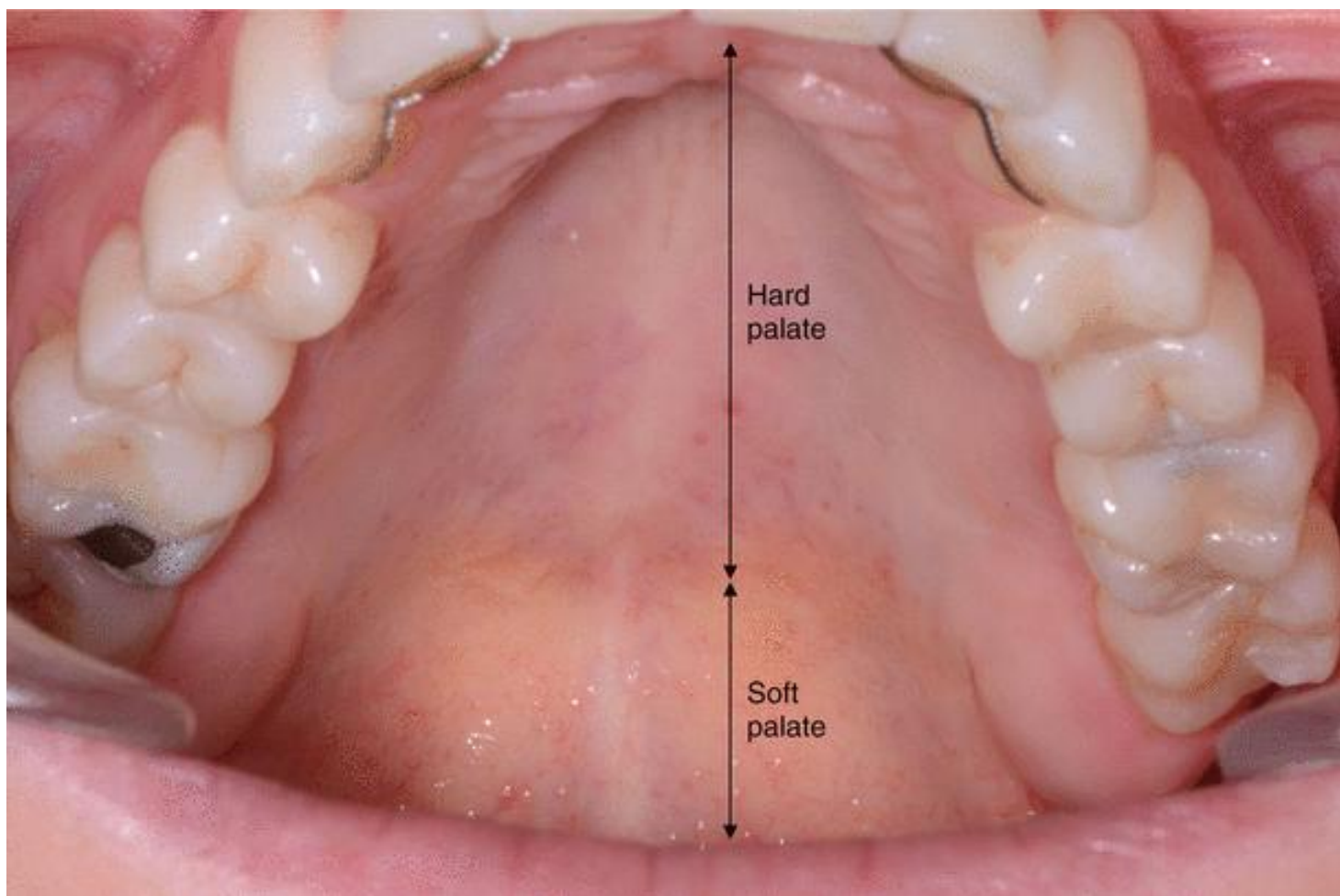


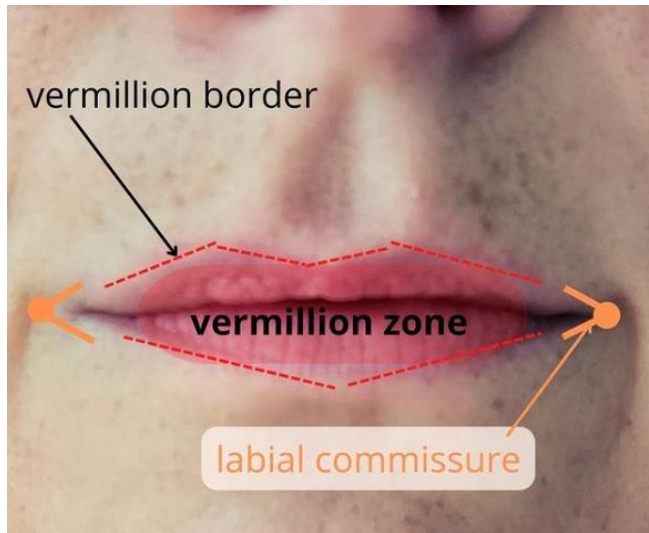
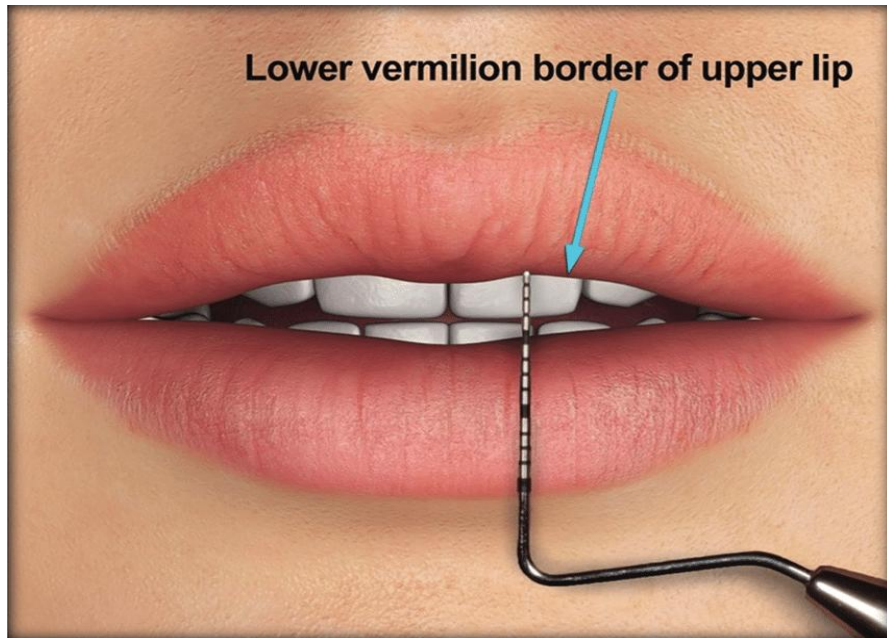
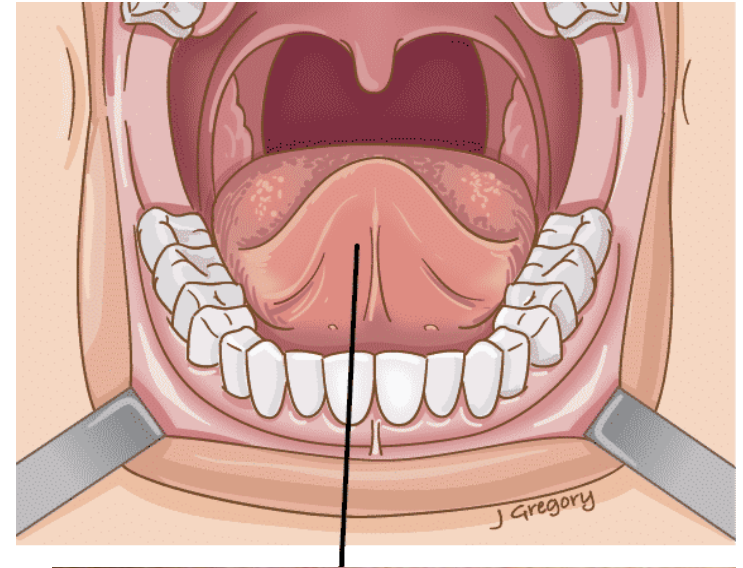
Organization of OMM

- Superiorly: hard and soft palate
- Inferiorly: floor of mouth, base of tongue
- Posteriorly: pillars of fauces, tonsils

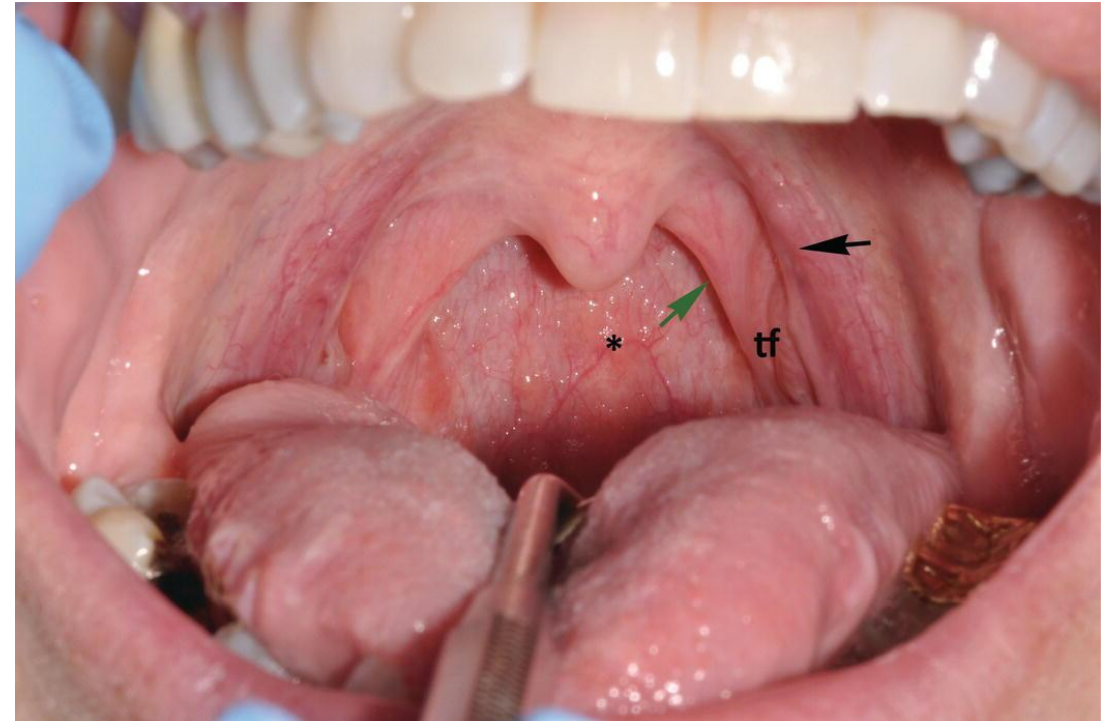
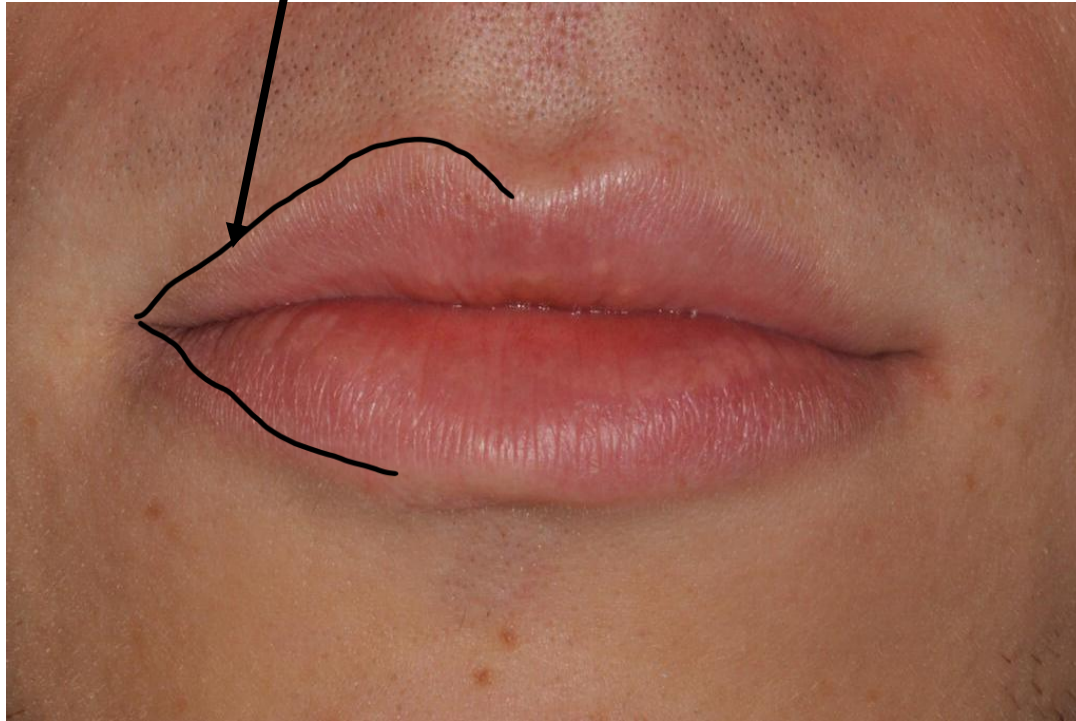








The oral mucosa is continuous with the skin of the lip through the vermilion border. Posteriorly it is continuous with the mucosa of the pharynx.



Classification of the Oral Mucosa

Based upon Primary functional criteria,

- 1. Masticatory Mucosa (25%) :** Gingiva, Hard palate
- 2. Lining Mucosa (covers 60% of total area):** Lip, cheek, vestibular fornix, alveolar mucosa, floor of mouth and soft palate
- 3. Specialized (sensory) Mucosa (15%):** Dorsum of the tongue and taste buds)

Masticatory Mucosa:

- **25% of total mucosa.**
- The masticatory mucosa is **bound to bone and does not stretch.** It bears forces generated when food is chewed.
- **Gingiva and hard palate.**
- **Keratinized** stratified squamous epithelium.

Lining Mucosa:

- **60% of total mucosa.**
- Covers the **floor of mouth, ventral(underside)tongue, alveolar mucosa, cheeks, lips and soft palate.**
- It covers the **musculature and is distensible**, adapting itself to the contraction and relaxation of cheeks, lips, and tongue
- **Non-keratinized** stratified squamous epithelium.
- Soft, moist, ability to stretch and compress

Specialized Mucosa:

- **Dorsum of tongue and taste buds**
- It is so called because it bears the **taste buds, which have a sensory function.**

Based upon Keratinization

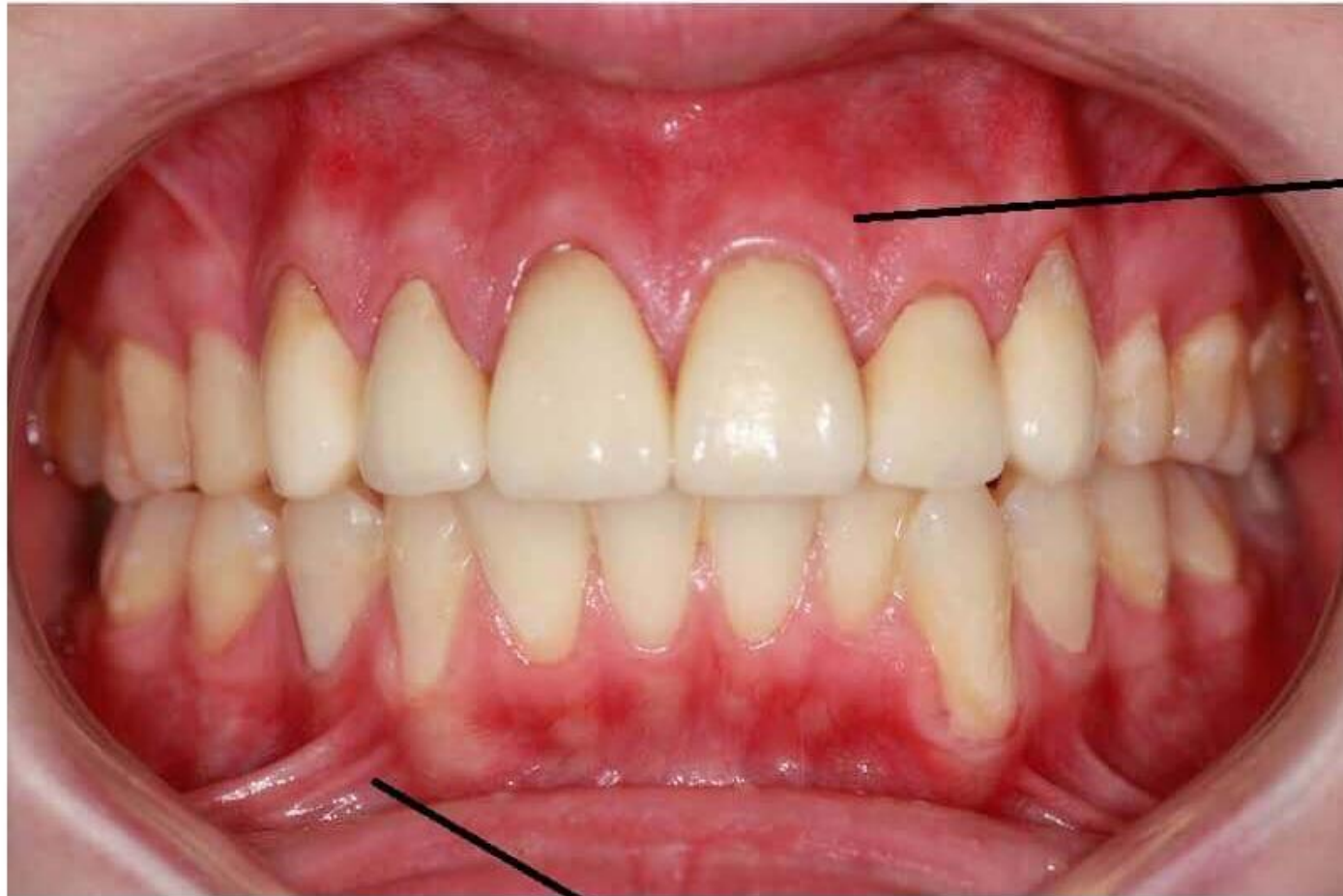
1. Keratinized mucosa-Gingiva, Hard palate and Vermilion

Boarder of lip.

- Orthokeratinized

- Parakeratinized

2. Non-keratinized mucosa



**Keratinized
Mucosa**

Non-Keratinized Mucosa

Based upon Location:

- Buccal Mucosa
- Lingual Mucosa
- Palatal Mucosa
- Labial Mucosa
- Alveolar Mucosa

Functions of oral mucous membrane (PASS)

1. **Protection:** Barrier for mechanical trauma and microbiological insults.
2. **Sensation:** Temperature (heat and cold), touch, pain, reflexes such as swallowing, gagging and salivation.
3. **Absorption:** Nitrates are absorbed sublingually.
4. **Secretion:** Salivary secretion creates a moist oral cavity in speech, swallowing, mastication and in the perception of taste.



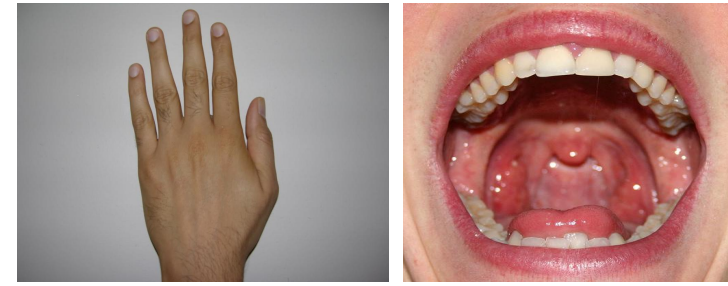
General features of oral mucous membrane

Separated from the skin by vermillion zone of the lips which is more deeply colored than rest of the oral mucosa.

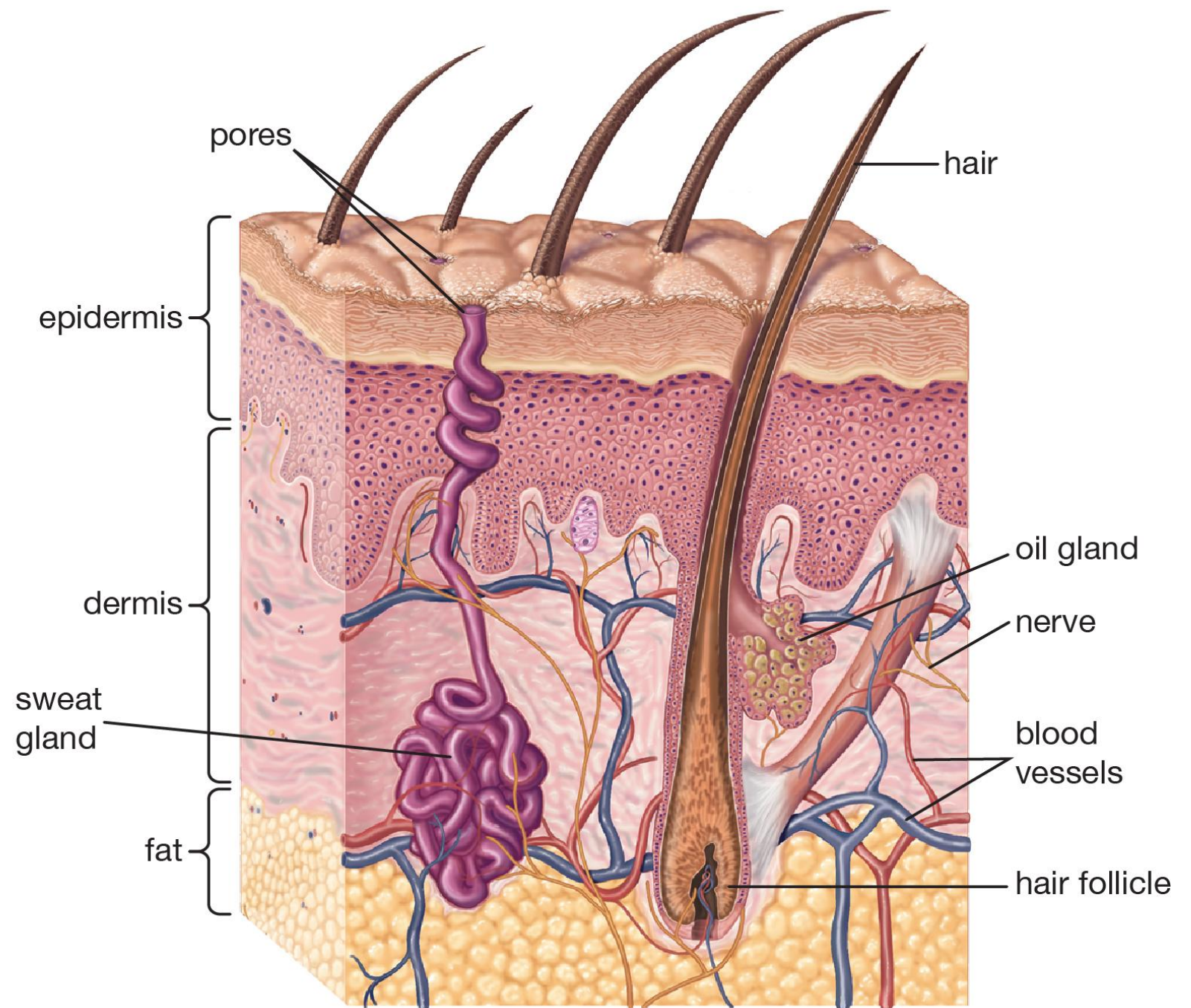
Factors affecting color of the oral mucosa:

- a. Concentration and state of dilation of the blood vessels in underlying connective tissue
- b. Thickness of the epithelium
- c. Degree of keratinization
- d. Amount of melanin pigmentation
- e. Clinically, color of oral mucosa is very important. For example, inflamed oral tissues appear red rather than the normal pale pink

How is the skin different from oral mucous membrane??



1. **Color** : More deeply coloured
2. **The epithelium** of the **skin is always orthokeratinized**, but the OMM depending on the region; may be nonkeratinized and if keratinized, the keratinization may be ortho or parakeratinized.
3. **Moist surface**: Presence of minor salivary glands in oral mucosa
4. **Absence of hair follicles, sweat gland and sebaceous gland.**
5. **Fordyce's spots**: Sebaceous glands in oral cavity predominantly in upper lip, buccal mucosa and alveolar mucosa
6. **Texture of surface**: Oral mucosa is smoother than the skin
7. **Firmness**: Oral mucosa varies in its firmness. For example buccal mucosa and lips are loose and pliable whereas the gingiva and hard palate are firm so critical clinically while giving injections



Basic structure of oral mucous membrane:

The epithelium of the oral mucous membrane is of the **stratified squamous** variety.

A stratified epithelium: more than one layer of epithelial cells.
topmost layer is made up of squamous (flattened and scale-like)

1. Overlying oral epithelium

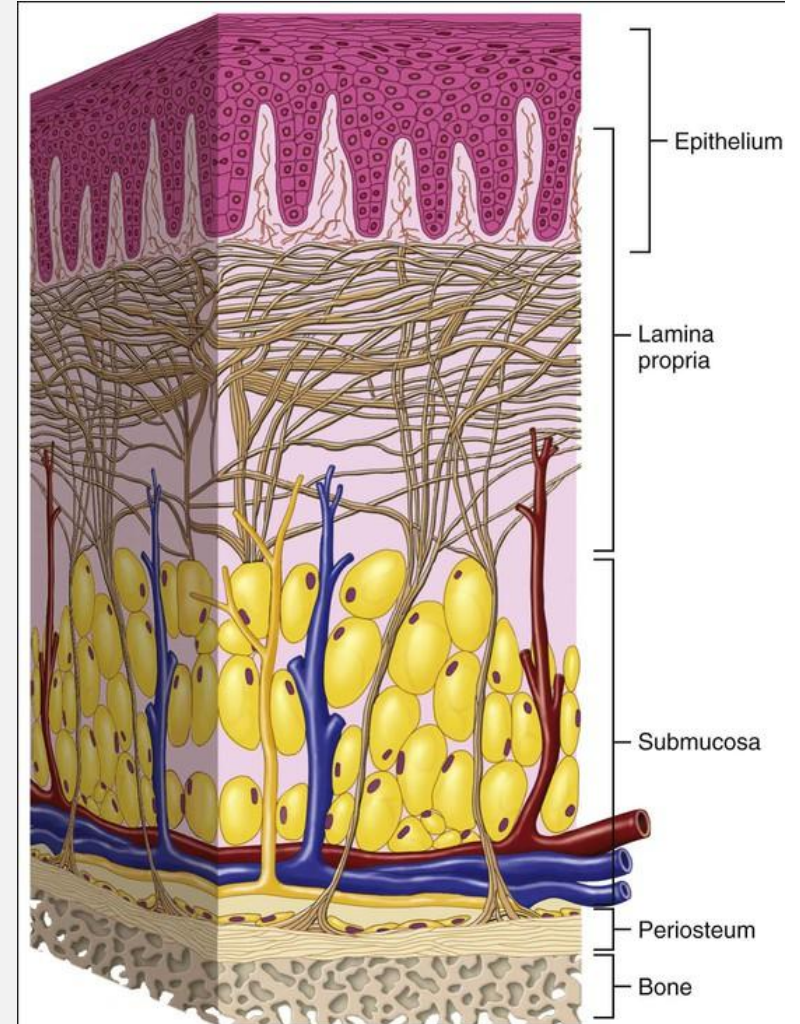
2. Underlying connective tissue

CT of oral mucus membrane: lamina propria

CT of skin: dermis

3. Submucosa

[Skin : Epidermis + Dermis]



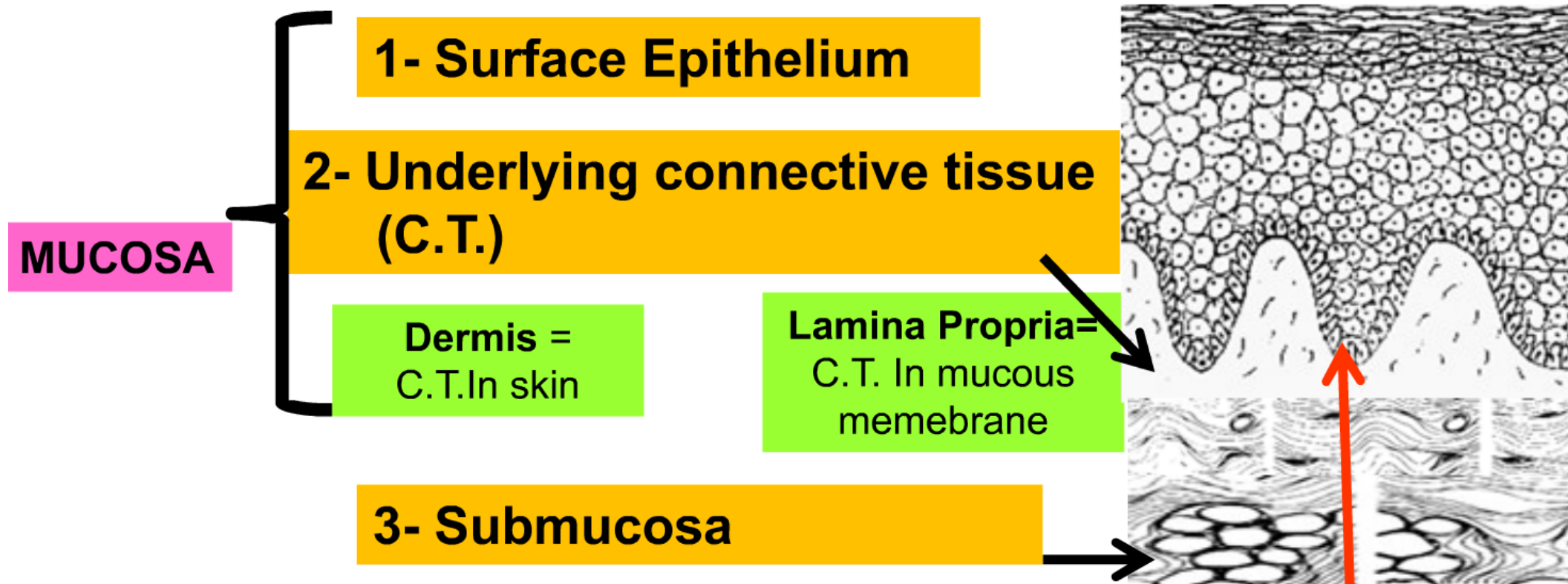
OVERLYING ORAL EPITHELIUM

- Made up of **stratified squamous epithelium**.
- The interface between epithelium and connective tissue is comprised of a structureless layer called **basement membrane**.
- This interface is irregular and is composed of downward projections of epithelium called **rete ridges or rete pegs**.

LAMINA PROPRIA

- Composed of collagen fibers, blood vessels, fibroblasts, endothelial cells, inflammatory cells, nerves and ground substance.
- It has 2 layers
 - a. **Superficial papillary layer** – irregularly arranged, loose and thin collagen fibres
 - b. **Deep reticular layer** – thick bundles of collagen fibres parallel to surface

Structure of oral mucous membrane



The epithelium is separated from the underlying lamina propria by **BASEMENT MEMBRANE**

Downward projections of the epithelium in the underlying C.T. is called rete-pegs or rete-ridges

1-Mucosa

2-Submucosa

- Present between lamina propria and underlying bone or muscles.
- Contain large bl.vs. & nerves, salivary glands, fat cells.....



A-Epithelium

St. Squamous epith.

B-Lamina propria (C.T.)

i-Superficial Papillary layer

- Increase the contact of epith surface with underlying vessels and nerves
- Composed of thin collagen fibers loosely arranged.

ii-Deep Reticular layer

- Contains deeper plexus of vessels & nerves
- Composed of thick bundles of collagen Arranged parallel to the surface

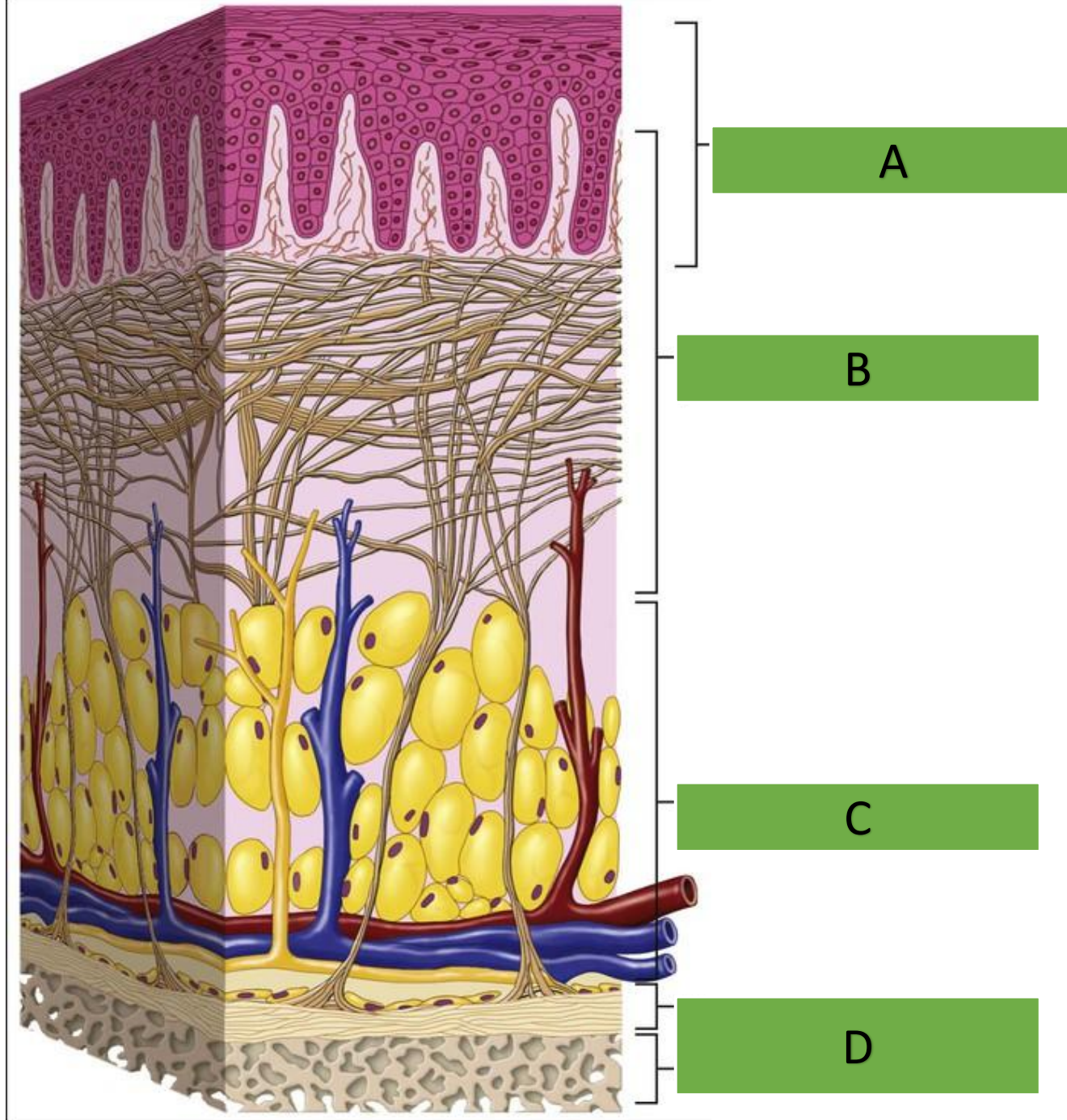
SUBMUCOSA

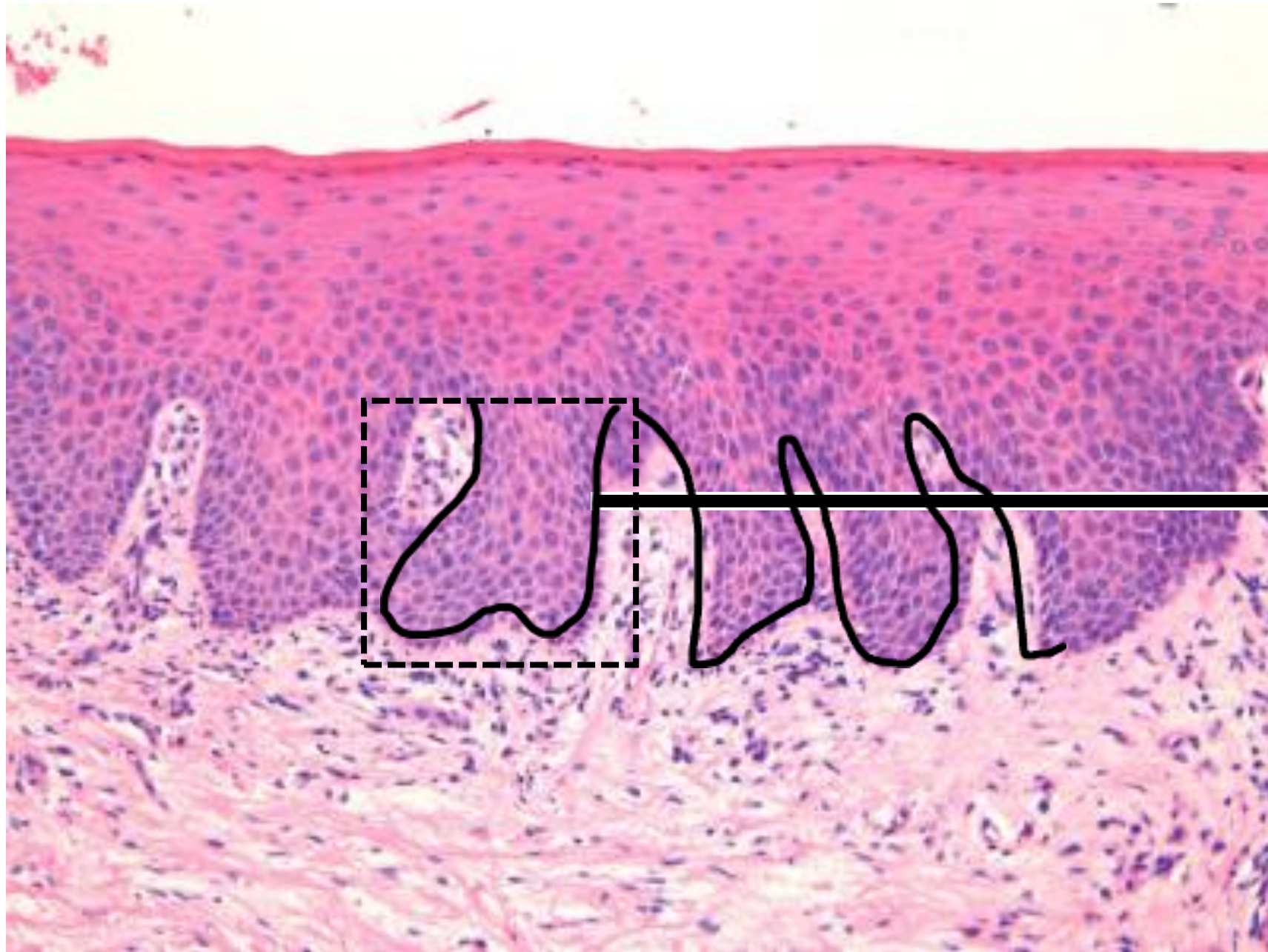
- Present between lamina propria and underlying bone or muscles.
- Loose fat and glandular tissue with large blood vessels and nerves seen in submucosa - this layer provides **flexibility**.
- Whether this attachment is loose or firm depends on the character of the submucosa.
- In gingiva and hard palate, **no submucosa** is seen and the lamina propria is directly attached to the periosteum of the underlying bone which provides **firm, inelastic attachment** –this is called **oral mucoperiosteum**.



**RAISE
YOUR
HAND**

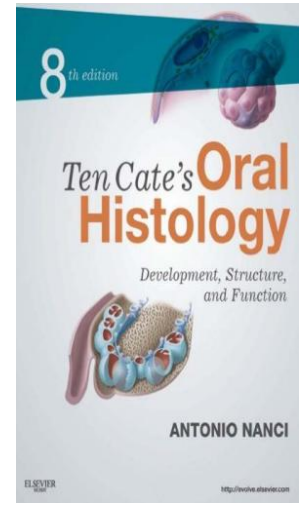
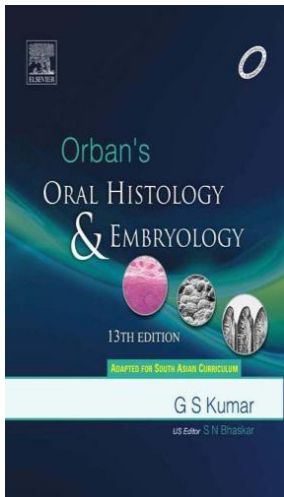






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- Orban's Oral Histology and Embryology Bhaskar, s. N. 11th edition.
- Oral Histology : Development, Structure and Funct Tencate, a. R. 4th edition.



Thank
you!!!
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